Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

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***Abstract*—The present report outlines a classical content-based image retrieval (CBIR) system using computer vision on multi-label aerial imagery. The AID multi-label dataset consists of 3,000 images, which include 2,400 images for training and 600 images for testing. Handcrafted HSV color, LBP texture, and HOG shape features are utilized in the present CBIR system. Feature-weighting, spatial features, multi-scale LBP, retrievals, and reliability estimation were applied in the course of the project to improve the baseline system. The proposed V2 model involves HSV, two LBP scale, and HOG. The present model scored Class P@10 of 0.3867, ML P@10 of 0.3650, and nDCG@10 of 0.6740 on the test set.**

***Index Terms*—CBIR, aerial images, multi-label retrieval, HSV, LBP, HOG, image retrieval, reliability**

I. Introduction

Content-based image retrieval (CBIR) searches for images by means of their visual content rather than by just searching for keywords or filenames. The goal of this project is the creation of a simple and understandable CBIR system for multi-label aerial images. The system finds visually similar images and gives a user information about how reliable the search results are.

The dataset that was used for this project is AID Multi-Label Dataset. The work was conducted gradually over four weeks. In Week 1, classical feature baseline was developed. In Week 2, feature fusion and spatial features were implemented. In Week 3, multi-scale LBP was added. In Week 4, retrieval explanation and reliability were included.

II. Dataset and Method

The database has 3,000 aerial images, 2,400 training images, and 600 test images. Training images form the retrieval gallery, whereas test images form the query during final evaluation.

The pipeline has feature extraction, similarity measurement, ranking, evaluation, and explanation. The three major handcrafted features include HSV, which is for color information; LBP, which is for local texture; and HOG, which is for shape and edge information.

Spatial information was considered in Week 2, where a 2×2 grid was considered. In Week 3, another setup for LBP was introduced with radius 2 and 16 sampling points. The final V2 model had feature weights as follows:

$$S = 0.10S_{HSV} + 0.35S_{LBP1} + 0.35S_{LBP2} + 0.20S_{HOG}. \quad (1)$$

The V2 weights were selected using five-fold validation on the training set. The test set was also used earlier in development (eight Week 1 baselines and V1 were evaluated on it), so the final results are a comparison across developed models rather than a fully blind holdout estimate.Multi-label relevance was defined as Jaccard 0.8 between the 17-label annotation vectors.

TABLE I
Dataset and Final Model Configuration

Item	Value
Total images	3,000
Training / test	2,400 / 600
Image size	600×600
HSV weight	0.10
LBP-$r1/p8$ weight	0.35
LBP-$r2/p16$ weight	0.35
HOG weight	0.20

III. Experimental Progress

Week 1 – Baseline: HSV, LBP, and HOG features were individually tested, then merged together. The most optimal baseline had a ratio of 0.50/0.25/0.25 for HSV/LBP/HOG and attained Class P@10 = 0.3345 and ML P@10 = 0.3483.

Week 2 – Feature Fusion: Weight tuning via validation was presented. For the optimal V1 model, HSV = 0.10, LBP = 0.70, and HOG = 0.20 were used, and it scored 0.4396 and 0.6894 on the single validation split (Class P@10 and nDCG@10 respectively). The effect of spatial HSV and LBP features was also explored.

Week 3 – Multi-scale LBP: Four types of retrieval methods were tested through five fold validation. Multi-scale LBP ranked first on the composite selection score in all five folds. Its mean Class P@10 was 0.4336 versus 0.4288 for V1, a small gain (+0.0048, 95

Week 4 – Explanation and Reliability: Component-level similarity evidence (percentile rank of each descriptor) and a reliability metric were added to enhance the interpretation of the retrieval results. In the reliability analysis, top-1 similarity was used to distinguish retrievals with high, medium, and low



Fig. 1. Week 2 retrieval examples using spatial features and difficult retrieval cases.

reliability. The final cutoffs selected were 0.930 for LOW and 0.960 for HIGH.

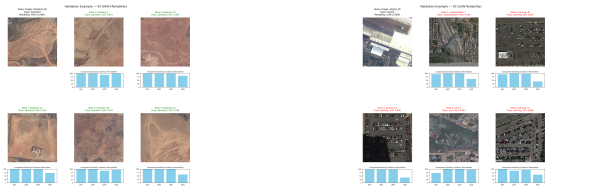


Fig. 2. Examples of high- and low-reliability retrievals from Week 4.

IV. Final Test Results

The V2 model was fixed before final testing. The final evaluation was performed using 600 test queries. Table II compares the baseline, V1, and final V2 model.

TABLE II
Final Test Results

Model	C-P@5	ML-P@5	C-P@10	ML-P@10	nDCG@10
Week 1 Best	0.3687	0.3603	0.3345	0.3483	0.6527
V1	0.4187	0.3803	0.3733	0.3593	0.6699
V2 (Final)	0.4183	0.3857	**0.3867**	**0.3650**	**0.6740**

Compared with the Week 1 baseline, the final V2 model improves Class P@10 from 0.3345 to 0.3867 and ML P@10 from 0.3483 to 0.3650. It also achieves Jaccard@10 of 0.6491, Class mAP of 0.1889, and ML mAP of 0.2278.

The final-test examples include two strong retrievals and one difficult retrieval. These examples show both the strengths and limitations of the final model.



Fig. 3. Final-test examples: strong forest retrieval, strong parking retrieval, and difficult river retrieval.

V. Limitations and Conclusion

With handcrafted global features used in the system, the complex objects and their relations are not considered fully. The class-based results might differ too because of 20 test queries per class used. Reliability is a rather primitive measure and does not guarantee the correctness of retrieval.

Overall, the project developed a classical computer-vision based CBIR system through four stages. The final V2 model was frozen before the final evaluation. Multi-scale LBP provided the best validation variant, and Week 4 introduced explanations and reliability measures into it. As a result, the final V2 model demonstrated Class P@10 = 0.3867, ML P@10 = 0.3650, and nDCG@10 = 0.6740. Handcrafted features prove to be effective for retrieval; however, visually similar aerial scenes still pose difficulty.